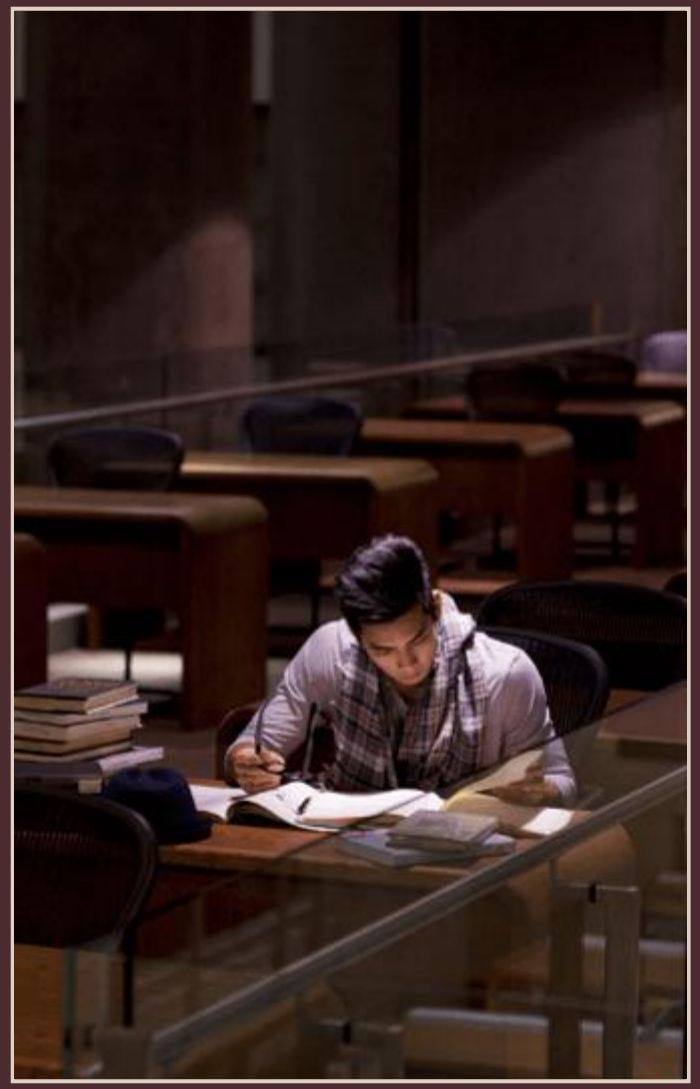
MILESTONE 4

Merin Kuriakose

- **The Challenge:** Companies struggle to process thousands of text-based reviews across multiple products.
- **The Solution:** An automated pipeline that performs:
- **Sentiment Analysis:** Categorizing tone.
- **Confidence Scoring:** Measuring AI certainty.
- **Keyword Extraction:** Identifying recurring themes.
- **Objective:** To provide a real-time, filterable view of product health.

The Problem & Objective



Key Performance Metrics

•**Total Reviews Analyzed**: 5000 individual customer feedbacks were processed.
•**Positive Sentiment**: Dominates the dataset at **54.1%** (2,705 reviews).
•**Negative Sentiment**: Accounts for **30.6%** (1,532 reviews), identifying significant areas for potential product improvement.
•**Neutral Sentiment**: Represents **15.3%** (763 reviews) of the feedback.

Long-Term Sentiment Trends

- **Sentiment Distribution**: The visual data confirms a healthy majority of positive feedback (green), while the negative feedback (red) is roughly half the volume of the positive.
- **Confidence Score Distribution**: The histogram shows a strong peak, indicating that the AI model is highly confident in its sentiment classifications for a large portion of the reviews.
- **Product Insights**: Filters are currently active for high-profile products including **Acer Predator**, **Amazon Fire**, **Anker Power**, and **Apple AirPods**, allowing for granular cross-product analysis.

Model Reliability

- The **Confidence Score Distribution** validates the accuracy of our sentiment analysis.
- **High Confidence:** The histogram shows a significant concentration of scores near the 0.0 to 0.75 range, indicating the AI model is highly certain of its categorizations for the bulk of the data.
- **Model Integrity:** Very few reviews fall into the "uncertain" or low-confidence ranges (near -1.0 or 1.0 extremes in this specific visualization context), ensuring that the dashboard's insights are actionable.

Sentiment Trends Over Time (2024–2025)

1. **Dominant Positive Sentiment:** Monthly positive reviews (green line) consistently lead the dataset, generally staying between 130 and 170 reviews.
2. **Critical Spike:** A significant surge in negative sentiment occurred in **April 2025**, reaching its peak for the entire period.
3. **Neutral Baseline:** Neutral feedback remains the most stable, typically hovering around 40–50 reviews monthly.
4. **Slide 3: Product-Specifi**

Product-Specific Health (Heatmap)

Identify which products are driving our reputation and which are posing risks.

- **High Performers:** Products like the **HP Omen 16** (52 positive reviews) and **Asus RT-AX88U** (51 positive reviews) show strong customer loyalty.
- **Risk Areas:** Products such as the **Garmin Venu 2** (20 negative reviews) show higher-than-average dissatisfaction density compared to top performers.
- **Strategic Action:** Direct resources toward improving the product features or support for items showing higher red-zone density in the heatmap.

Deep Dive: Keyword Insights

Primary Success Drivers

"Satisfied," "Happy," and "Quality" are the most frequent positive descriptors..

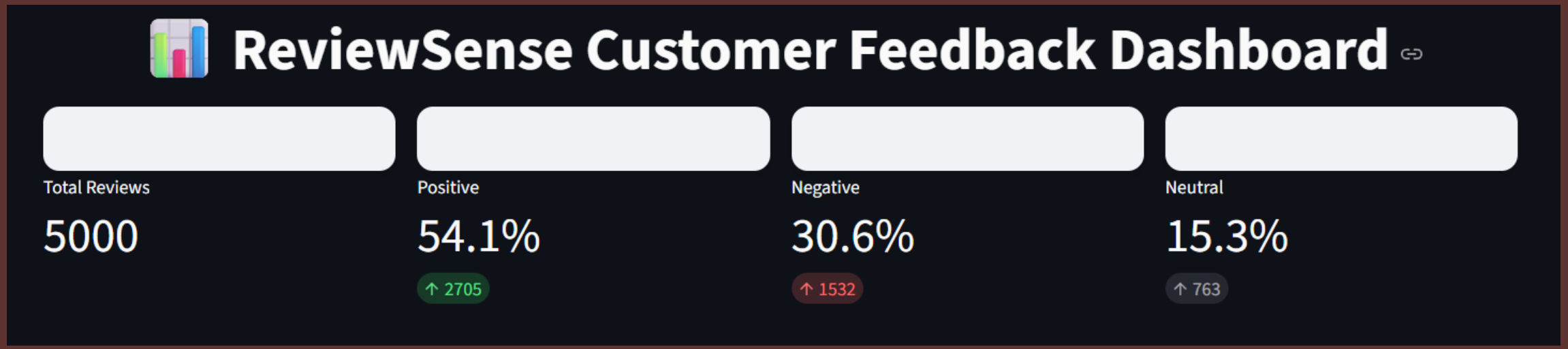
Key Pain Points

Negative feedback is characterized by terms like "Dissatisfied," "Disappointed," and concerns regarding "Battery" and "Build"..

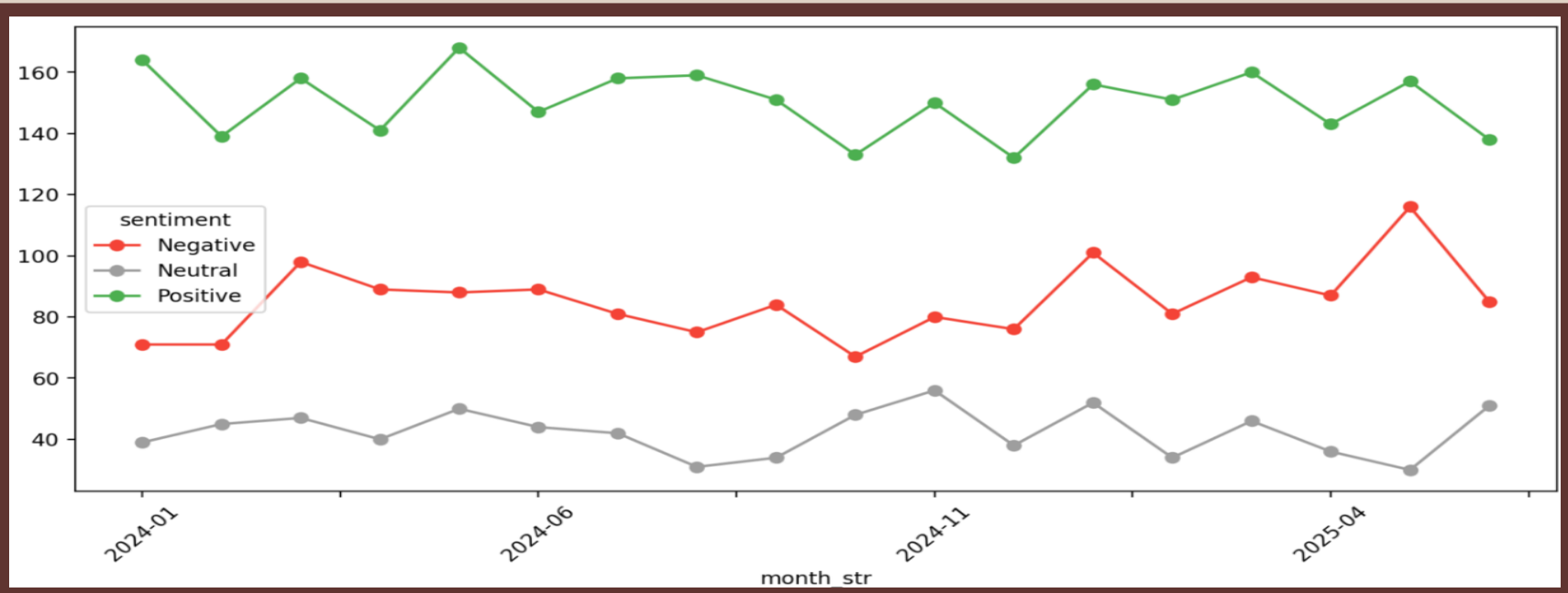
Actionable Word Cloud

Visualizing these keywords shows that while "Performance" is a highly mentioned strength, "Software" and "Experience" are areas where customers feel mixed.

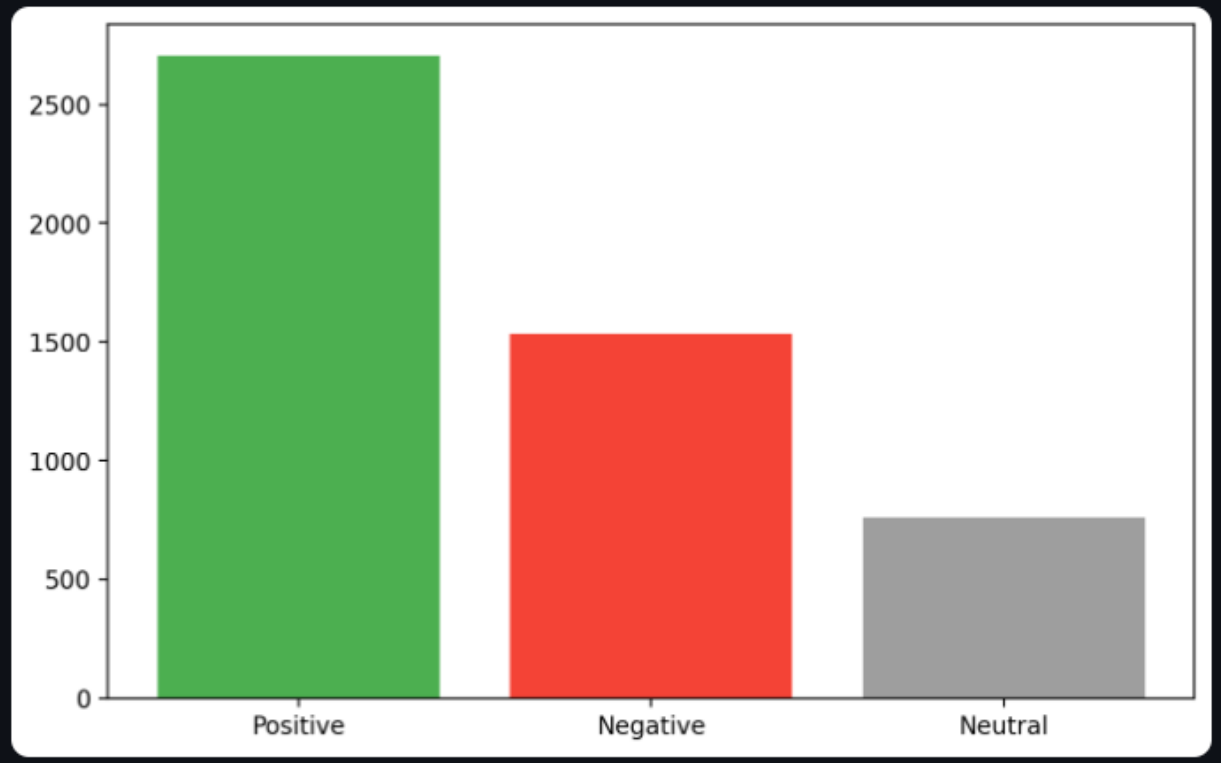
OUTPUT


ReviewSense Customer Feedback Dashboard
Total Reviews
5000
Positive
54.1%
↑ 2705
Negative
30.6%
↑ 1532
Neutral
15.3%
↑ 763

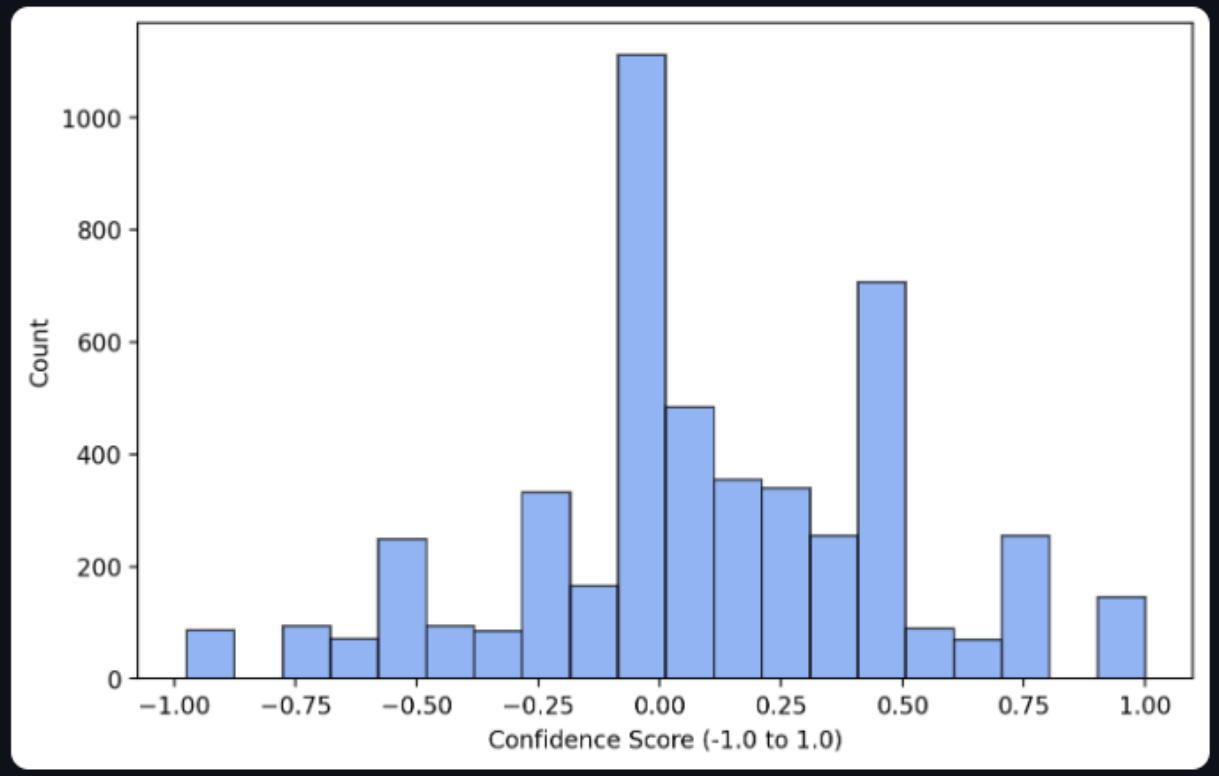
Sentiment Trends Over Time (2024–2025)



😊 Sentiment Distribution

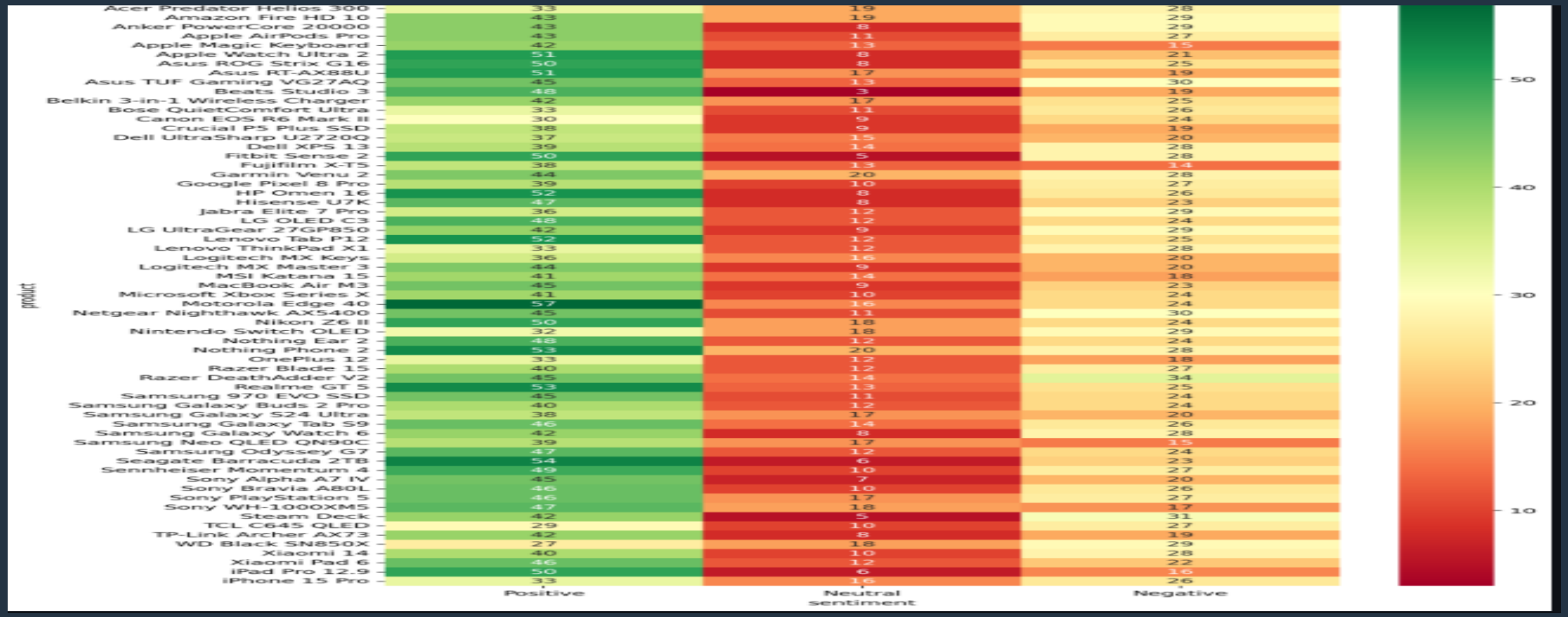


🎯 Confidence Score Distribution



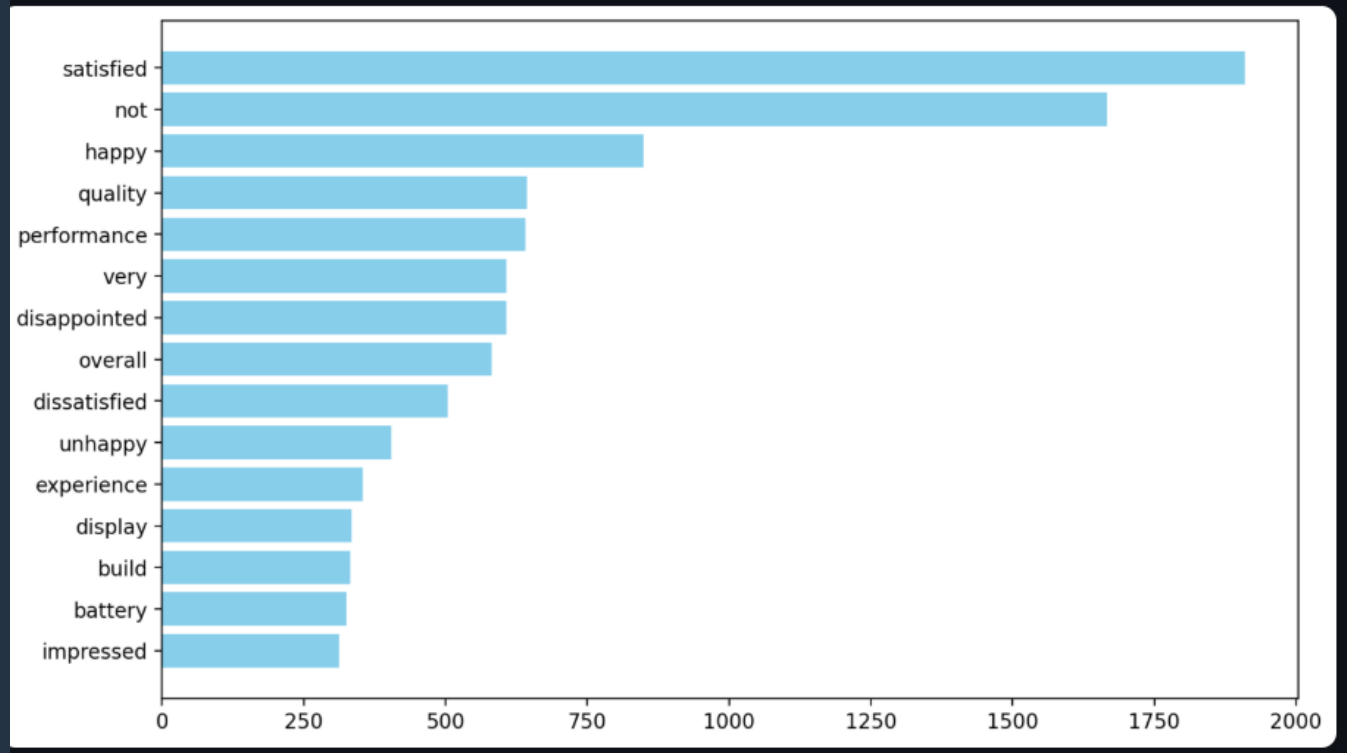
SENTIMENT AND CONFIDENCE SCORE

HEATMAP

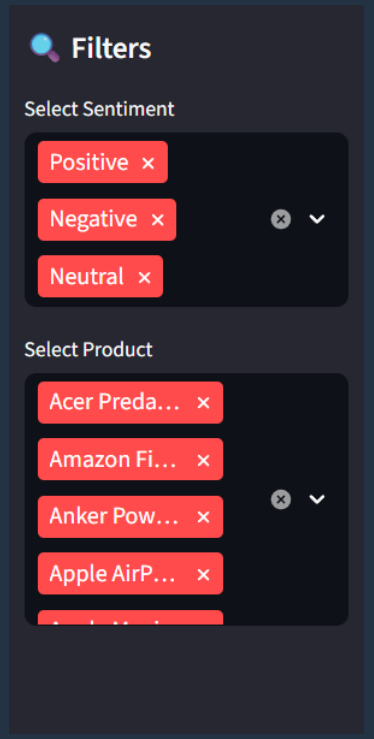

product
Acer Predator Helios 300 33 19 28
Amazon Fire HD 10 43 19 29
Anker PowerCore 20000 43 8 29
Apple AirPods Pro 43 11 27
Apple Magic Keyboard 42 13 15
Apple Watch Ultra 2 51 8 21
Asus ROG Strix G16 50 8 25
Asus RT-AX88U 51 17 19
Asus TUF Gaming VG27AQ 45 13 30
Beats Studio 3 48 3 19
Belkin 3-in-1 Wireless Charger 42 17 25
Bose QuietComfort Ultra 33 11 26
Canon EOS R6 Mark II 30 9 24
Crucial P5 Plus SSD 38 9 19
Dell UltraSharp U2720Q 37 15 20
Dell XPS 13 39 14 28
Fitbit Sense 2 50 5 28
Fujifilm X-T5 38 13 14
Garmin Venu 2 44 20 28
Google Pixel 8 Pro 39 10 27
HP Omen 16 52 8 26
Hisense U7K 47 8 23
Jabra Elite 7 Pro 36 12 29
LG OLED C3 48 12 24
LG UltraGear 27GP850 42 9 29
Lenovo Tab P12 52 12 25
Lenovo ThinkPad X1 33 12 28
Logitech MX Keys 36 16 20
Logitech MX Master 3 44 9 20
MSI Katana 15 41 14 18
MacBook Air M3 45 9 23
Microsoft Xbox Series X 41 10 24
Motorola Edge 40 57 16 24
Netgear Nighthawk AX5400 45 11 30
Nikon Z6 II 50 18 24
Nintendo Switch OLED 32 18 29
Nothing Ear 2 48 12 24
Nothing Phone 2 53 20 28
OnePlus 12 33 12 18
Razer Blade 15 40 12 27
Razer DeathAdder V2 45 14 34
Realme GT 5 53 13 25
Samsung 970 EVO SSD 45 11 24
Samsung Galaxy Buds 2 Pro 40 12 24
Samsung Galaxy S24 Ultra 38 17 20
Samsung Galaxy Tab S9 46 14 26
Samsung Galaxy Watch 6 42 8 28
Samsung Neo QLED QN90C 39 17 15
Samsung Odyssey G7 47 12 24
Seagate Barracuda 2TB 54 6 23
Sennheiser Momentum 4 49 10 27
Sony Alpha A7 IV 45 7 20
Sony Bravia A80L 46 10 26
Sony PlayStation 5 46 17 27
Sony WH-1000XM5 47 18 17
Steam Deck 42 5 31
TCL C645 QLED 29 10 27
TP-Link Archer AX73 42 8 19
WD Black SN850X 27 18 29
Xiaomi 14 40 10 28
Xiaomi Pad 6 46 12 22
iPad Pro 12.9 50 6 16
iPhone 15 Pro 33 16 26
Positive
Neutral sentiment
Negative
50
40
30
20
10

KEYWORD INSIGHTS

🔑 Keyword Insights



FITERS AND DOWNLOADING OPTION



THANK YOU